

6.4.1.1

$$a) \operatorname{Re}(\langle u, v \rangle) = \frac{1}{4}(\|u+v\|^2 - \|u-v\|^2)$$

$$b) \operatorname{Im}(\langle u, v \rangle) = \frac{1}{4}(\|u-iv\|^2 - \|u+iv\|^2)$$

$$a) \frac{1}{4}(\|u+v\|^2 - \|u-v\|^2) = \frac{1}{4}(\langle u+v, u+v \rangle - \langle u-v, u-v \rangle)$$

tohle vyplývá z vlastnosti skalárního součinu $\langle u, u \rangle = \|u\|^2$

pak z linearity $\langle \cdot, \cdot \rangle$

$$= \frac{1}{4}(\langle u, u \rangle + \langle u, v \rangle + \langle v, u \rangle + \langle v, v \rangle - (\langle u, u \rangle - \langle u, v \rangle - \langle v, u \rangle + \langle v, v \rangle))$$

$$= \frac{1}{4}(\langle u, v \rangle + \langle v, u \rangle + \langle u, v \rangle + \langle v, u \rangle)$$

pak ze ~~skalar~~ Hermitovskosti:

$$= \frac{1}{4}(\langle u, v \rangle + \overline{\langle u, v \rangle} + \langle u, v \rangle + \overline{\langle u, v \rangle})$$

$$= \frac{1}{2}(\langle u, v \rangle + \overline{\langle u, v \rangle})$$

$\langle u, v \rangle$ je komplexní číslo, a pro $x \in \mathbb{C}$ platí $x + \overline{x} = 2\operatorname{Re}(x)$

(přesně $a+bi + (a-bi) = 2a$)

$$\text{tedy} = \frac{1}{2}(2\operatorname{Re}(\langle u, v \rangle) = \operatorname{Re}(\langle u, v \rangle))$$

$$b) \text{opět} \approx \langle u, u \rangle = \|u\|^2$$

$$\frac{1}{4}(\langle u-iv, u-iv \rangle - \langle u+iv, u+iv \rangle)$$

$$= \frac{1}{4}(\langle u, u \rangle + \langle u, iv \rangle - \langle iv, u \rangle + \langle iv, iv \rangle - (\langle u, u \rangle + \langle u, iv \rangle + \langle iv, u \rangle + \langle iv, iv \rangle))$$

$$= \frac{1}{4}(-\langle u, iv \rangle - \langle u, iv \rangle - \langle u, iv \rangle - \langle u, iv \rangle)$$

$$= \frac{1}{2}(-\langle u, iv \rangle - \overline{\langle u, iv \rangle}) = \frac{1}{2}(-i\langle u, v \rangle - i\overline{\langle u, v \rangle})$$

$$= \frac{1}{2}(-i\langle u, v \rangle + i\overline{\langle u, v \rangle}) = \frac{1}{2}(-i(\langle u, v \rangle - \overline{\langle u, v \rangle}))$$

~~tože~~ $x - \overline{x} = 2i\operatorname{Im}(x)$

$$\text{nebo } \langle u, v \rangle = a+bi, a, b \in \mathbb{R}, \text{ pak} = \frac{1}{2}(-i(a+bi - a+bi)) = \frac{1}{2}(-i(2bi)) = b = \operatorname{Im}(\langle u, v \rangle)$$